DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 6

# Data mining specialist

Profession 6 of 10 in Development and IT.

## What the profession means

A data mining specialist finds hidden patterns in large datasets: groups of customers, unusual transactions and associations between products.

## What professionals do every day

Explores data, applies clustering, association and anomaly methods, validates findings and explains them to the business.

## Why businesses hire them, and what problems they solve

Patterns the business does not expect can reveal new segments and risks.

- Patterns that are chance, not real.
- Results nobody can act on.
- Data that is too messy to mine.

## Skills you need

- Exploratory analysis.
- Clustering and segmentation.
- Association rules and anomaly detection.
- Validation on new data.

## Tools (free and paid)

| Tool | What it is for | Free option | Paid option | Official link (not checked live) |
|---|---|---|---|---|
| Python and pandas | Analysis in code | Free | Free | pandas.pydata.org |
| scikit-learn | Clustering and models | Free | Free | scikit-learn.org |
| SQLite | Learn SQL on a local file | Free | Free | www.sqlite.org |

## Learning sequence: beginner, intermediate, advanced

| Level | What you can do when you finish |
|---|---|
| **Beginner** | Segment customers with three rules. |
| **Intermediate** | Find unusual orders with a simple rule. |
| **Advanced** | Validate a segment on a second month. |

## Projects by level

- **Beginner:** Customer segmentation for a shop.
- **Intermediate:** Fraud-style anomaly check for refunds.

- **Advanced:** Product affinity analysis for a bakery.

## Client-ready portfolio project

A customer segmentation for a fictional shop with three segments, their sizes and spend, and a validation note.

## Worked example

**Worked example.** Customers with spend of 2,000 or more form the high segment; the segment is checked on the next month before the marketing team uses it.

## Assignment with full solution

ASSIGNMENT

### Spend segments with validation

**Business problem:** A shop wants three customer groups for campaigns.

**Steps**

1. Run the segmentation query on the sample data.
2. Count the customers and total spend in each segment.
3. Check the segments on a second period.
4. Write a note on stability.

**Expected output:** The segment table and the validation note.

SOLUTION (ANSWER KEY)

SQL

```
-- simple customer segment by total spend
SELECT c.name,
       SUM(o.amount) AS spend,
       CASE WHEN SUM(o.amount) >= 2000 THEN 'high'
            WHEN SUM(o.amount) >= 1000 THEN 'medium' ELSE 'low' END AS segment
FROM customers c JOIN orders o ON o.customer_id = c.id
GROUP BY c.id, c.name
ORDER BY spend DESC;
```

Results from the query: Ravi 4,500 (high); Asha 2,700 (high); Meera 2,000 (high, exactly on the threshold); Tara 1,500 (medium); Imran 400 (low). Meera sits on the 2,000 boundary, so a small change in one order moves her segment: that is the stability question to report. Re-run on another period before using the segments for campaigns.

## Common mistakes

- Reporting segments without checking they are stable.
- Choosing thresholds after seeing the results.
- Ignoring privacy rules when grouping customers.
- No business owner for the segments.

## Freelancing opportunities

- Segmentation for marketing teams.
- Anomaly checks for finance and operations.
- Briefs for 'customer segmentation', 'data mining', 'clustering'.

## Learning resources

**LEARNING RESOURCES (VERIFY BEFORE USE)**

- scikit-learn clustering guide

Part of **Master Freelancing: Data Analytics and Data Science**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.